COMMON ALLOYS

Bronze
The first man-made alloy, bronze was produced around 5,000 years ago by smelting copper and tin together. Bronze is resistant to atmospheric corrosion and is extremely strong.

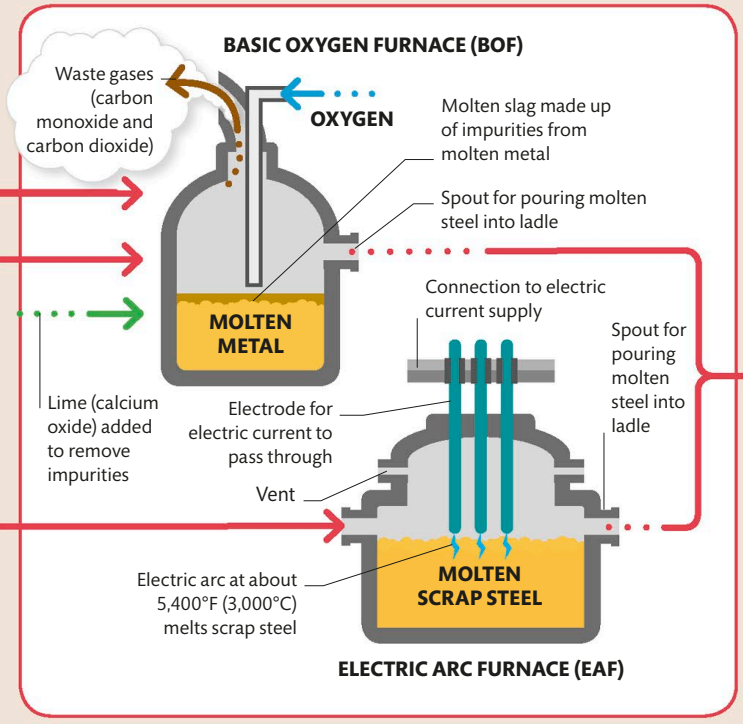
Sterling silver
Sterling silver is an alloy consisting of 92.5 percent silver and 7.5 percent other metals, most commonly copper. These other metals make sterling silver harder and stronger than pure silver.

Solder
Traditionally, solder was an alloy of tin and lead, but modern solder usually consists of tin, copper, and silver and typically melts between about 355°F (180°C) and 375°F (190°C).

Cast iron
Cast iron is an alloy of iron and carbon, with a carbon content greater than about 2 percent. It is easy to cast and has good corrosion resistance and excellent compressive strength.

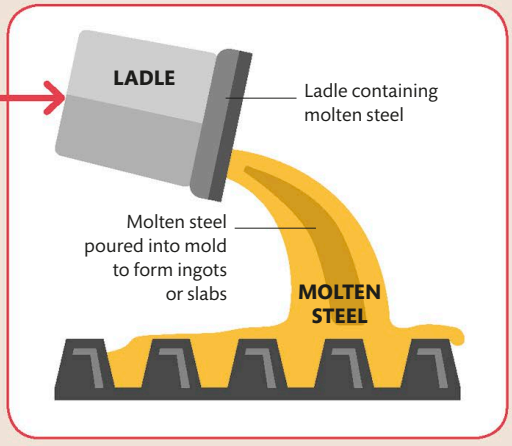
Brass
An alloy of copper and zinc, brass has a relatively low melting point (about 1,650°F/ 900°C), which makes it easy to cast. Brass is durable, more malleable than bronze, and has a bright, goldlike finish.

Stainless steel
Stainless steel varies in composition but commonly consists of 74 percent iron, 18 percent chromium, and 8 percent nickel. The chromium makes the alloy resistant to corrosion.



PURE **IRON** IS JUST **SOFT** ENOUGH TO **CUT** WITH A SHARP **KNIFE**





3 Making molten steel
In a basic oxygen furnace, oxygen is blown into molten pig iron, which reduces the iron's carbon content and produces steel. Lime is also added to remove impurities, which form a molten layer of slag. Sometimes, scrap steel may also be added. In an electric arc furnace, scrap steel is simply melted.

4 Casting or rolling molten steel
The molten steel may be poured into a ladle and then into a mold or put through rollers to shape it. This basic steel may be made into finished products, or it may be reprocessed by adding alloying elements to produce high-grade or special steels.